METAL-SEMICONDUCTOR JUNCTION

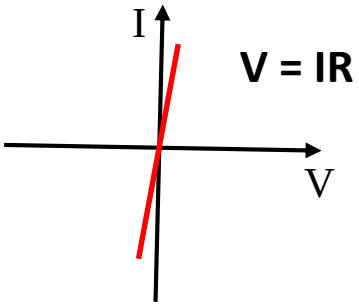
- Metal/semiconductor junctions
  - Possible variations
- Metal/n-semiconductor ($\phi_m > \phi_s$) junctions
- Other possible metal/semiconductor junctions
- Conditions for Ohmic or rectifying behaviour

Metal/semiconductor junctions

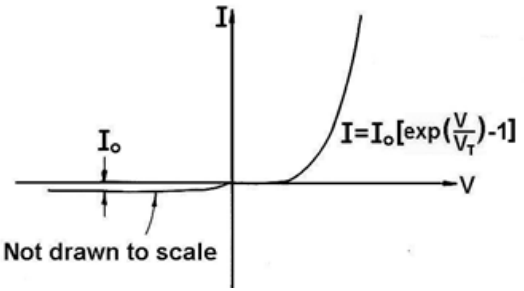
- Oldest practical semiconductor device
  - Using tungsten filament and galena (lead sulphide) crystal
  - Called cat's whisker detector
  - One of the earliest demonstrations was by Sir Jagadish Chandra Bose in 1901 as microwave detector
- Metal-semiconductor (M/S) junctions are integral parts of any semiconductor device
  - Used for electrical contacts
    - Need to be ohmic (equally conducting in both directions)
- M/S junctions can also be rectifying
  - Called Schottky diodes
  - Often used instead of p-n junctions due to some advantages
  - Can have lower forward drop due to higher reverse saturation current
- M/S junctions are **unipolar devices**
  - No delays associated with minority carrier storage
  - Used for high frequency operation
    - For example, microwave mixer circuits for frequency up-conversion or down-conversion

Types of metal/semiconductor junctions

- Ohmic or Schottky depends on
  - Work function difference between metal and semiconductor
  - n-type or p-type semiconductor
- Four possibilities for M/S junctions
  - n-type semiconductor, $\phi_m > \phi_s$
  - n-type semiconductor, $\phi_m < \phi_s$
  - p-type semiconductor, $\phi_m > \phi_s$
  - p-type semiconductor, $\phi_m < \phi_s$



Ohmic



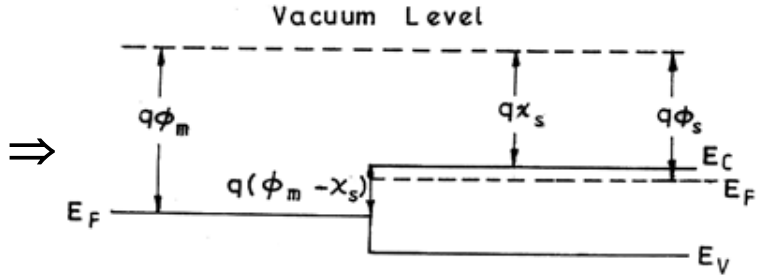
Schottky (rectifying)

Metal/n-type semiconductor ($\phi_m > \phi_s$) junction at thermal equilibrium

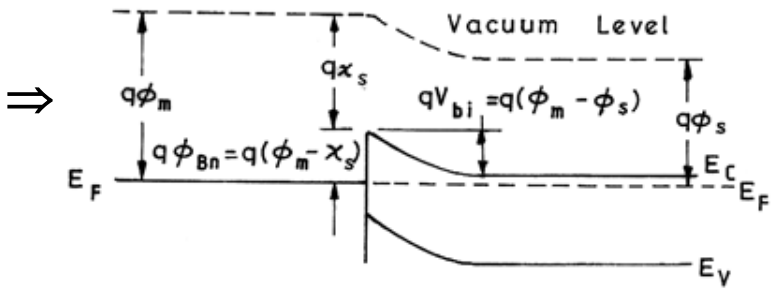
Constraints for drawing energy band diagram

1) There can be no discontinuity in vacuum level
2) Fermi levels must align at thermal equilibrium

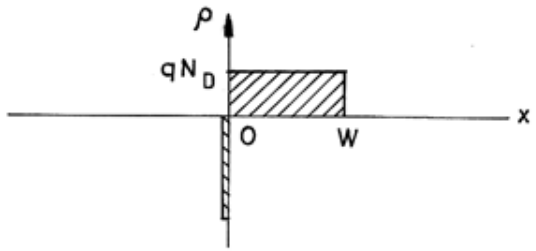
Energy band diagram of isolated metal and n-type semiconductor where $\phi_m > \phi_s$



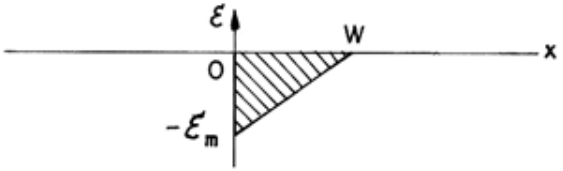
Energy band diagram of the metal/n-type semiconductor at thermal equilibrium



Charge distribution



Electric field profile



Definitions:

ϕ_{Bn} ⇒ Barrier height (or barrier potential) at heterojunction

V_{bi} ⇒ Built-in potential

W ⇒ Depletion layer width

$$\phi_{Bn} = \phi_m - \chi_s$$

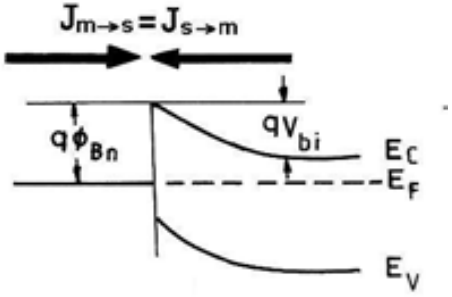
$$V_{bi} = \phi_m - \phi_s = \phi_{Bn} - (\phi_s - \chi_s)$$
$$= \phi_{Bn} - V_T ln\left(\frac{N_C}{N_D}\right)$$

$$W = \sqrt{\frac{2\varepsilon V_{bi}}{qN_D}}$$

$$E_m = \frac{qN_D W}{\varepsilon} = \frac{2V_{bi}}{W}$$

Similar to p^+n junction diode

Current flow at thermal equilibrium



- At thermal equilibrium
  - Electron flow from metal to semiconductor $(J_{m\to s})$
  - Electron flow from semiconductor to metal $(J_{s\to m})$
  - $\boldsymbol{J_{m\to s} = J_{s\to m}}$
- Dominant current flow mechanism
  - Thermionic emission
  - Electrons with sufficient energy overcome the barrier
  - Proportional to electron concentration at the interface (n_{s0})

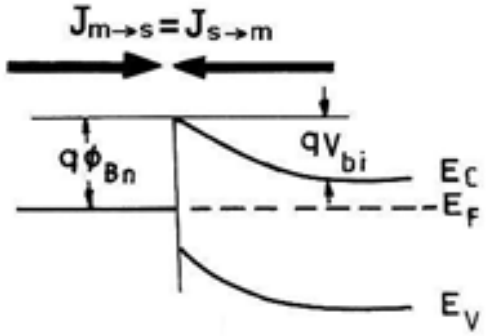
$$n_{s0} = N_C exp\left[-\frac{(E_C - E_F)|_{x=0}}{kT}\right] = N_C exp\left[-\frac{\phi_{Bn}}{V_T}\right] = N_D exp\left[-\frac{V_{bi}}{V_T}\right]$$

$$|J_{m\to s}| = |J_{s\to m}| = C_1 n_{s0} = C_1 N_C exp\left[-\frac{\phi_{Bn}}{V_T}\right] = A^{**}T^2 exp\left[-\frac{\phi_{Bn}}{V_T}\right]$$

$C_1 \Rightarrow$ proportionality constant $A^{**} \Rightarrow$ Richardson constant = 110 A/(K^{2}cm^2) for n-type Si

Current flow under bias

Thermal equilibrium
Equal flow of electrons in both directions

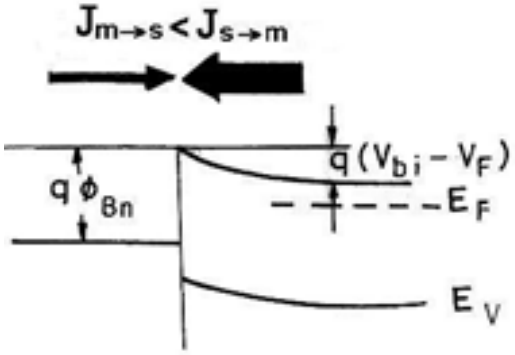


$$n_{s0} = N_D exp\left[-\frac{V_{bi}}{V_T}\right]$$

$$|J_{m\to s}| = |J_{s\to m}| = A^{**}T^2 exp\left[-\frac{\phi_{Bn}}{V_T}\right] = J_S$$

$$J = |J_{s\to m}| - |J_{m\to s}| = 0$$

Forward bias
Flow of electrons:
- metal to semiconductor unchanged
- semiconductor to metal increases exponentially

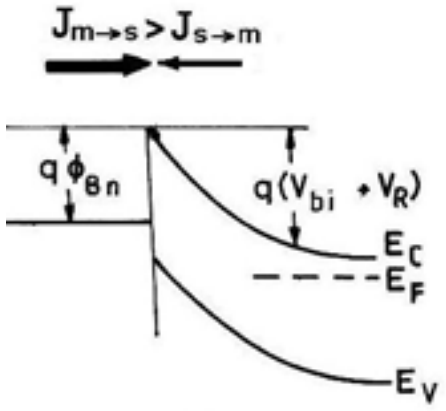


$$n_s = N_D exp\left[-\frac{(V_{bi} - V_F)}{V_T}\right] = n_{s0} exp\left[\frac{V_F}{V_T}\right]$$

$$|J_{m\to s}| = J_S \qquad |J_{s\to m}| = J_S exp\left[\frac{V_F}{V_T}\right]$$

$$J = |J_{s\to m}| - |J_{m\to s}| = J_S\left[exp\left(\frac{V_F}{V_T}\right) - 1\right]$$

Reverse bias
Flow of electrons:
- metal to semiconductor unchanged
- semiconductor to metal reduces



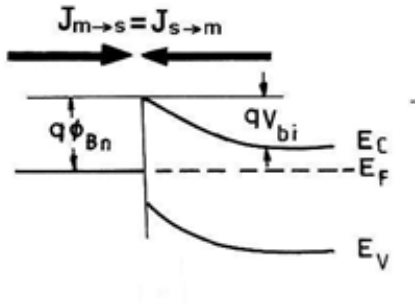
$$n_s = N_D exp\left[-\frac{(V_{bi} + V_R)}{V_T}\right] = n_{s0} exp\left[-\frac{V_R}{V_T}\right]$$

$$|J_{m\to s}| = J_S \qquad |J_{s\to m}| = J_S exp\left[-\frac{V_R}{V_T}\right]$$

$$J = |J_{s\to m}| - |J_{m\to s}| = J_S\left[exp\left(-\frac{V_R}{V_T}\right) - 1\right]$$

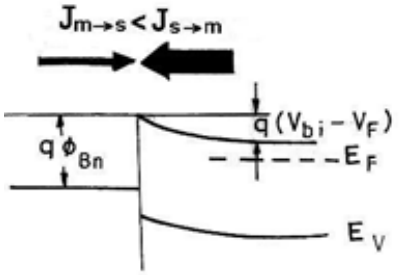
Unified expression for current

Thermal equilibrium



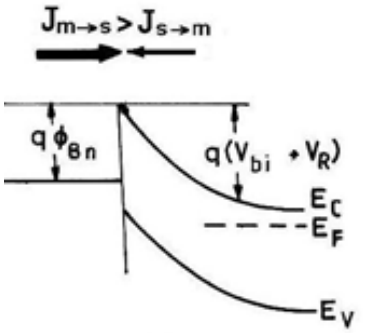
$$J = 0$$

Forward bias

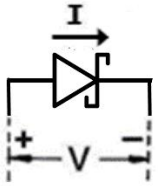


$$J = J_S\left[exp\left(\frac{V_F}{V_T}\right) - 1\right]$$

Reverse bias



$$J = J_S\left[exp\left(-\frac{V_R}{V_T}\right) - 1\right]$$

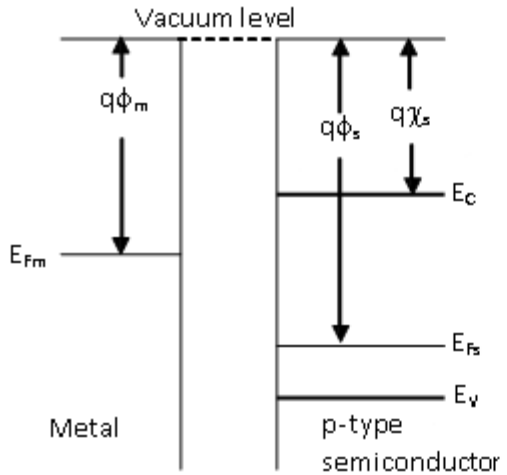


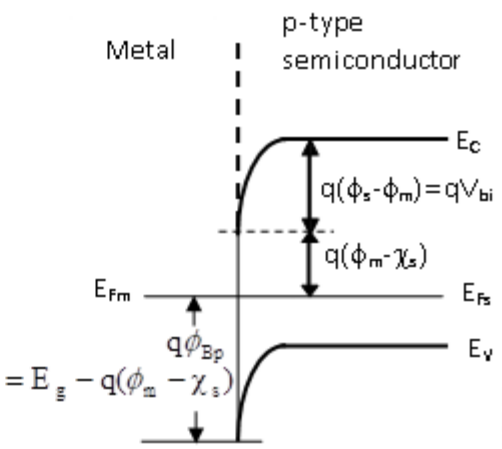
$$I = I_s\left[exp\left(\frac{V}{V_T}\right) - 1\right] \qquad I_s = AA^{**}T^2 exp\left[-\frac{\phi_{Bn}}{V_T}\right]$$

$A \Rightarrow$ Cross-sectional area

- Current expression in **Schottky diode** is exactly similar as in p-n junction diode
- Reverse saturation current is usually higher than in p-n junction diodes
  - Depends on $\phi_{Bn} = \phi_m - \chi_s$
  - Increases with temperature
- Only majority carriers involved
  - No delay associated with minority carrier storage
  - Suitable for very high frequency operation

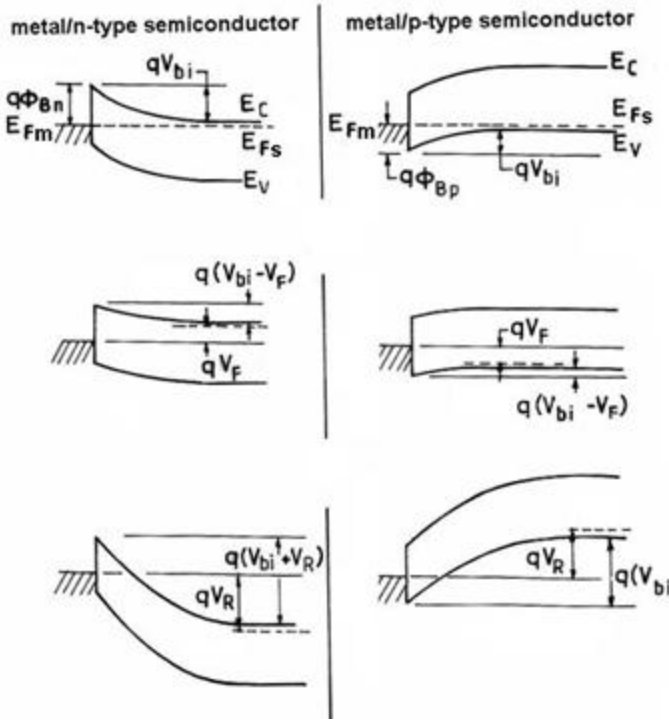
Metal/p-type semiconductor ($\phi_m < \phi_s$) junction





$$q\phi_{Bp} = E_G - q(\phi_m - \chi_s)$$

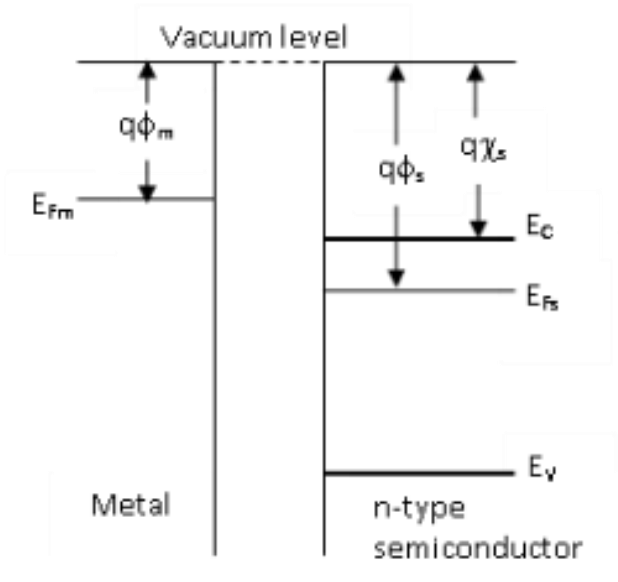
$$V_{bi} = \phi_s - \phi_m = \phi_{Bp} - (E_{Fs} - E_V)$$
$$= \phi_{Bp} - V_T ln\left(\frac{N_V}{N_A}\right)$$

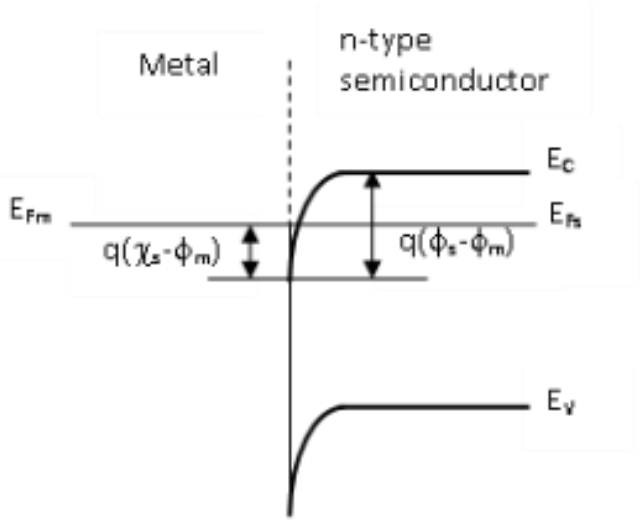


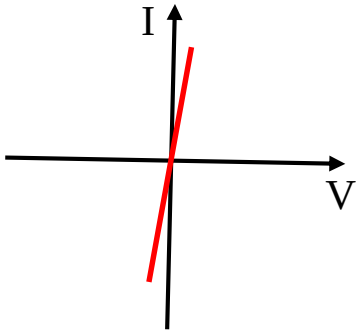
- Metal/p-type semiconductor ($\phi_m < \phi_s$) junction is a Schottky diode
  - **Behaves like n^+p junction**
  - At thermal equilibrium, hole emission current from metal to semiconductor ($|J_{m\to s}|$) is exactly equal to that from semiconductor to metal ($|J_{s\to m}|$)
  - $|J_{m\to s}|$is independent of bias
  - $|J_{s\to m}|$ is proportional to $exp\left(\frac{V}{V_T}\right)$
    - Increases at forward bias
    - Reduces at reverse bias

$$I = I_s\left[exp\left(\frac{V}{V_T}\right) - 1\right] \qquad I_s = AA^{**}T^2 exp\left[-\frac{\phi_{Bp}}{V_T}\right]$$

Metal/n-type semiconductor ($\phi_m < \phi_s$) junction

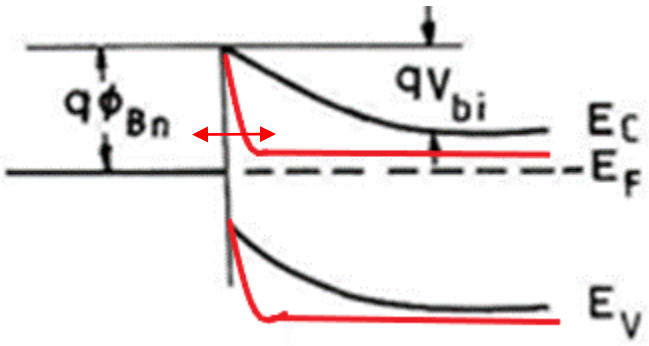






- Metal/n-type semiconductor ($\phi_m < \phi_s$) junction is ohmic
  - The semiconductor surface is accumulated with electrons
  - There is no barrier for electrons to flow from either side
- Metal/p-type semiconductor ($\phi_m > \phi_s$) junction is also ohmic

Metal/heavily doped semiconductor junctions



Metal/n⁺ semiconductor ($\phi_m > \phi_s$) junction at thermal equilibrium

- For heavily doped semiconductor, W is very low
  - Enables tunneling of electrons in both directions
    - Electrons can flow through the barrier
  - Junction shows ohmic behaviour

In summary

Semiconductor	Work functions	Type
n-type ($<10^{17}/cm^3$)	$\phi_m > \phi_s$	Schottky
n-type ($<10^{17}/cm^3$)	$\phi_m < \phi_s$	Ohmic
p-type ($<10^{17}/cm^3$)	$\phi_m > \phi_s$	Ohmic
p-type ($<10^{17}/cm^3$)	$\phi_m < \phi_s$	Schottky
n-type/p-type ($>10^{19}/cm^3$)	Does not matter	Ohmic

Some questions

- Why is metal/semiconductor junction suitable for high frequency operation?
- In Schottky diodes, why is $J_{m \to s}$ always constant, while $J_{s \to m}$ changes with bias
- Why is a junction of a metal and heavily doped semiconductor always ohmic?

THANK YOU

QUESTIONS???